

PHYS 708 — Assignment 5 (Due date: 24 November 2025)

Problem

(1)

(a) The spectral evolution of high-energy cosmic-ray electrons inside our Galaxy is governed by the following equation

$$\frac{dN(E)}{dt} = \frac{\partial}{\partial E}[b(E)N(E)] + Q(E)$$

where $N(E)$ is the cosmic-ray electron density, $b(E) = a E^2$ is the synchrotron and inverse Compton energy loss rate, and $Q(E) = \kappa E^{-p}$ is the source spectrum. For a continuous injection of electrons from sources uniformly distributed inside the whole Galactic volume, show that the spectrum inside the Galaxy at a given time t is:

$$N(E) = \begin{cases} \frac{\kappa E^{-(p+1)}}{a(p-1)} [1 - (1 - aEt)^{p-1}] & \text{if } E \leq E_0 \\ \frac{\kappa E^{-(p+1)}}{a(p-1)} & \text{if } E > E_0 \end{cases} ,$$

where $E_0 = 1/(at)$ is the energy at which the energy loss time becomes equal to time t .

[Hint: This solution is discussed in Slide 12 of Lecture 9. One way to solve the equation is to take Fourier transform in t . You can also follow any other methods you are familiar with. If you get stuck, please let me know].

(b) Analyse the shape of the electron spectrum at $E \leq E_0$ and $E > E_0$.